9A.BFS

```c
#include <stdio.h>
int graph[20][20], visited[20], n;

void BFS(int start){
    int queue[20], front=0, rear=0;
    visited[start]=1;
    queue[rear++]=start;
    while(front<rear){
        int node=queue[front++];
        printf("%d ", node);
        for(int i=0;i<n;i++){
            if(graph[node][i]==1 && !visited[i]){
                visited[i]=1;
                queue[rear++]=i;
            }
        }
    }
}

int main(){
    int start;
    printf("Enter number of vertices: ");
    scanf("%d",&n);
    printf("Enter adjacency matrix:\n");
    for(int i=0;i<n;i++)
        for(int j=0;j<n;j++)
            scanf("%d",&graph[i][j]);
    for(int i=0;i<n;i++)
        visited[i]=0;
    printf("Enter starting vertex: ");
    scanf("%d",&start);
    printf("BFS Traversal: ");
    BFS(start);
    return 0;
}
```

OUTPUT:

C:\Users\BMSCE\Desktop\1BF

```
Enter number of vertices: 4
Enter adjacency matrix:
0 1 1 0
1 0 0 1
1 0 0 1
0 1 1 0
Enter starting vertex: 2
BFS Traversal: 2 0 3 1
Process returned 0 (0x0)   execution time : 28.706 s
Press any key to continue.
```